

Northwind Apparel Co.

Demand Forecasting Analysis

Multi-Model Time Series Forecast
Jan 2023 – Dec 2025 + 12-Month Forward Projection

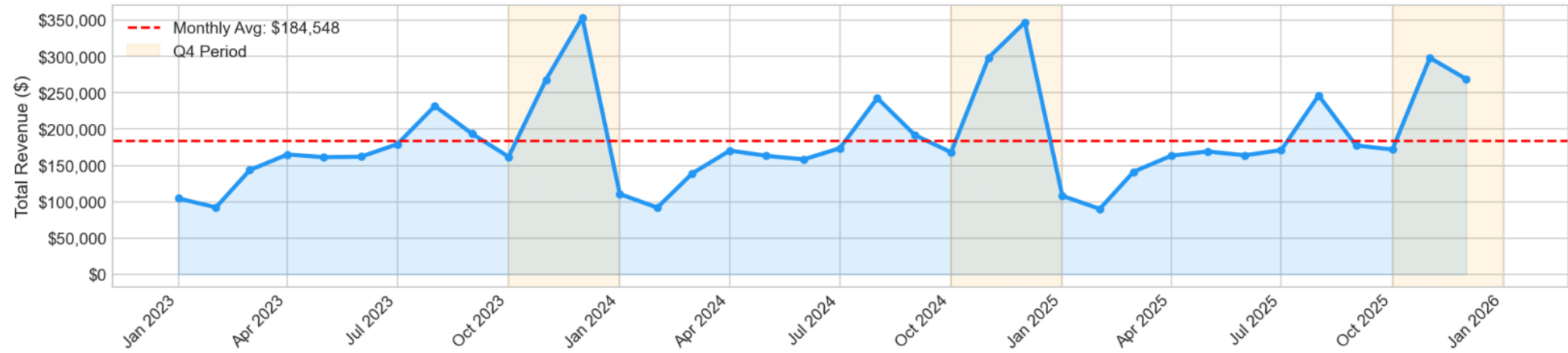
Model Performance Summary

SMA Baseline:	MAPE 37.11%
Holt-Winters:	MAPE 3.86%
Prophet (winner):	MAPE 2.73%

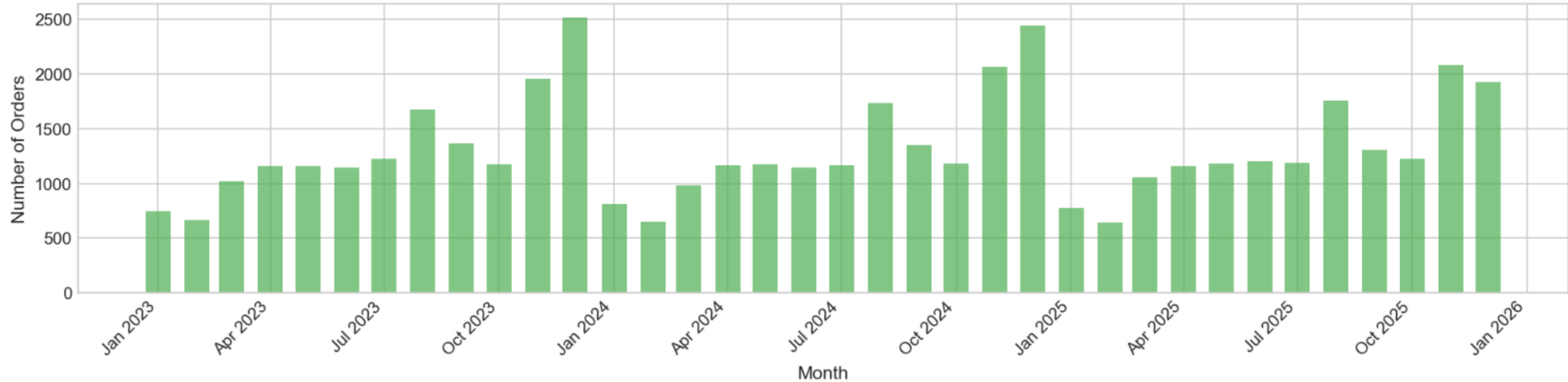
2026 Revenue Forecast:	\$1,966,619
YoY Change vs 2025:	-9.4%

EDA: Revenue & Order Volume Overview

Monthly Revenue — Northwind Apparel Co. (2023–2025)

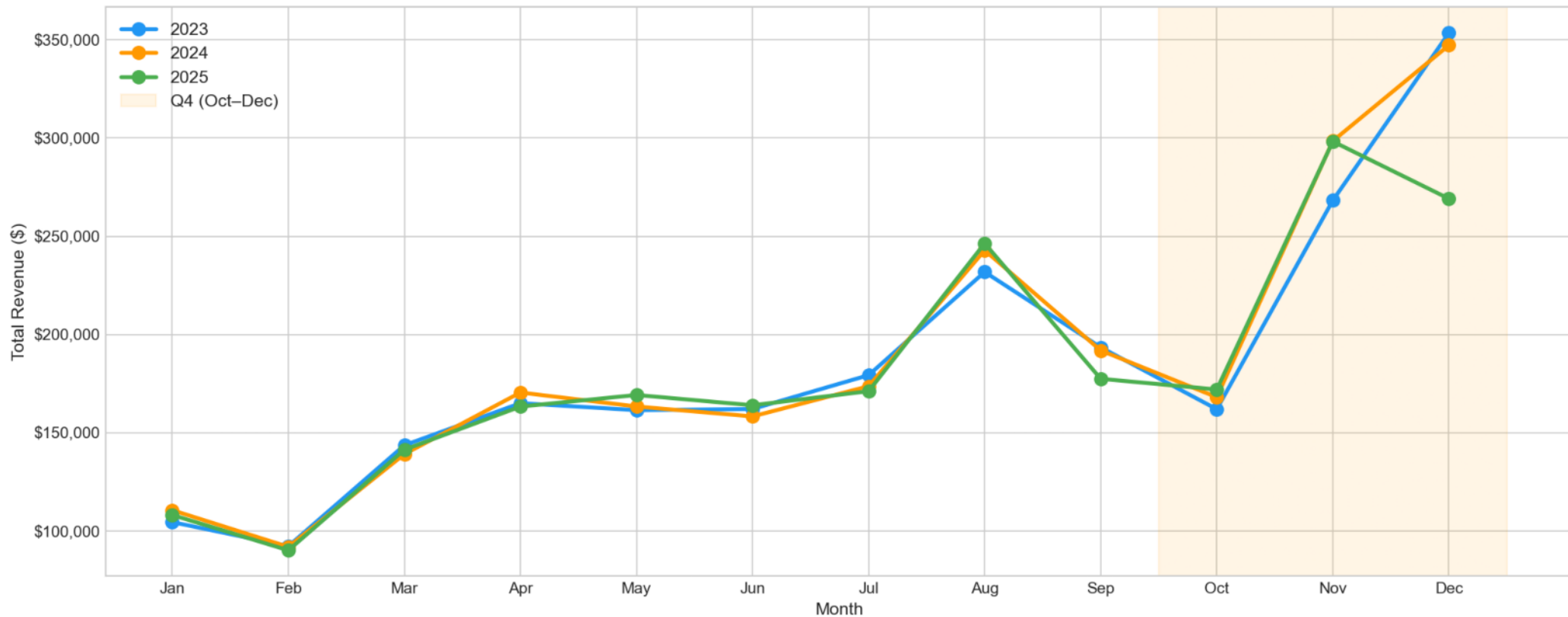


Monthly Order Count (2023–2025)



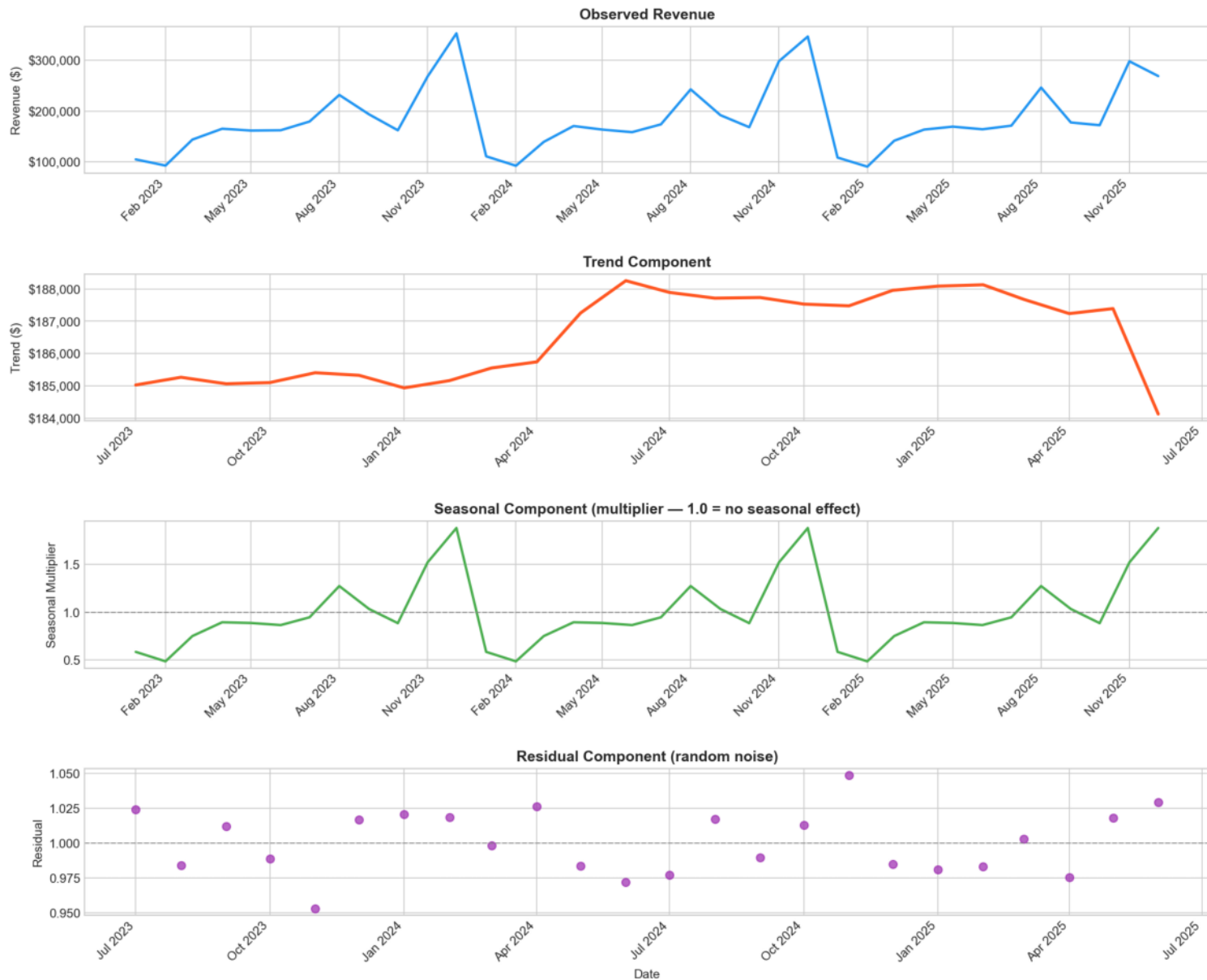
EDA: Seasonality by Year

Monthly Revenue by Year — Seasonality Pattern



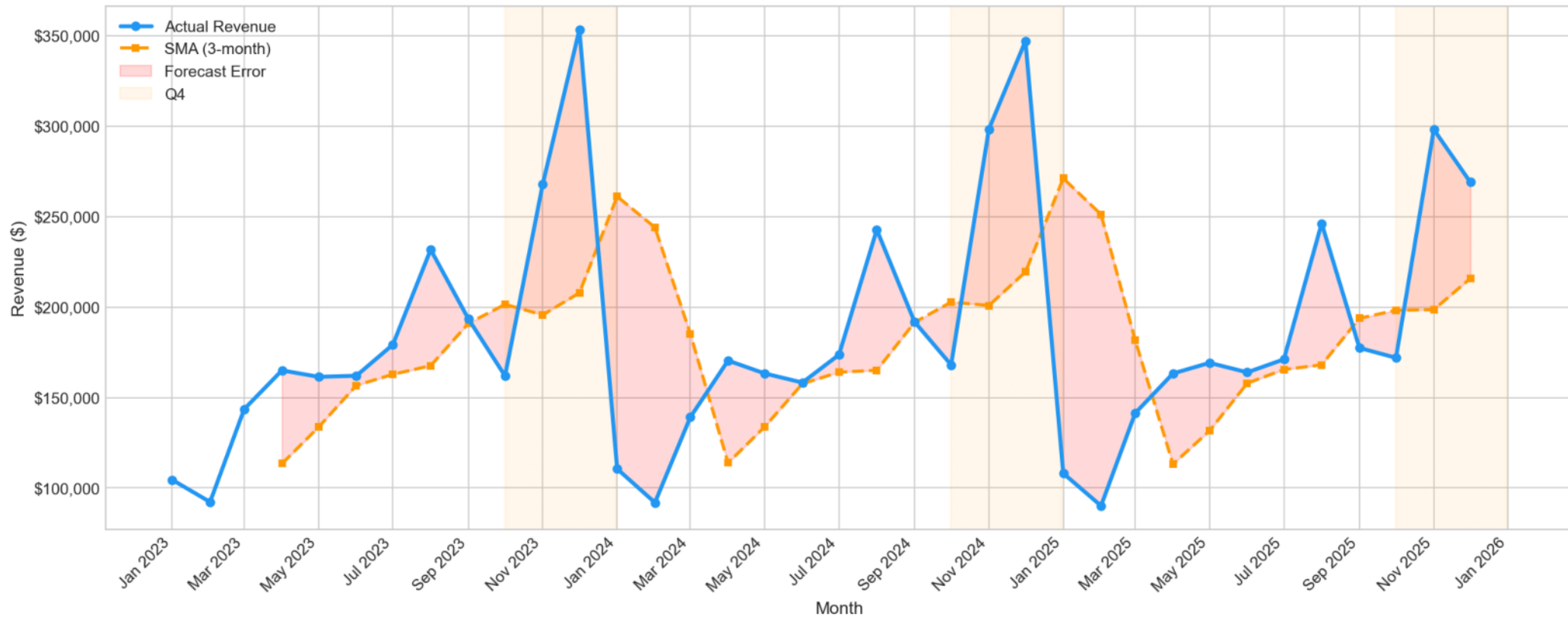
Time Series Decomposition

Time Series Decomposition — Northwind Apparel Revenue



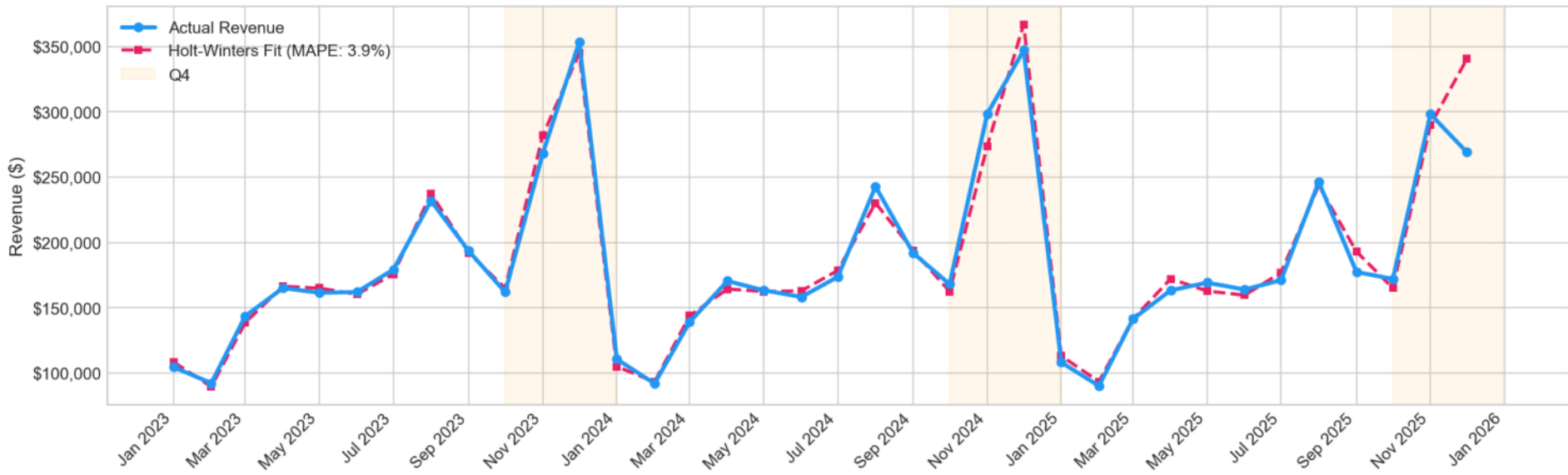
Model 1: Simple Moving Average (Baseline)

Simple Moving Average Forecast (MAPE: 37.1%) — Baseline Model

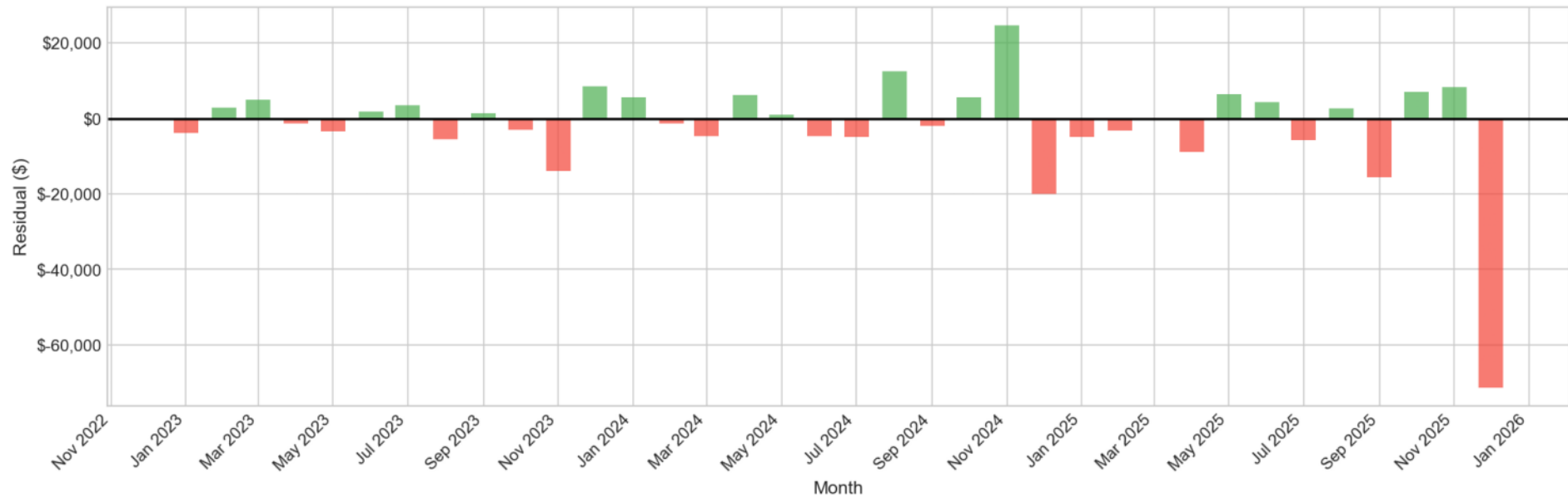


Model 2: Holt-Winters Exponential Smoothing

Holt-Winters Exponential Smoothing — Historical Fit

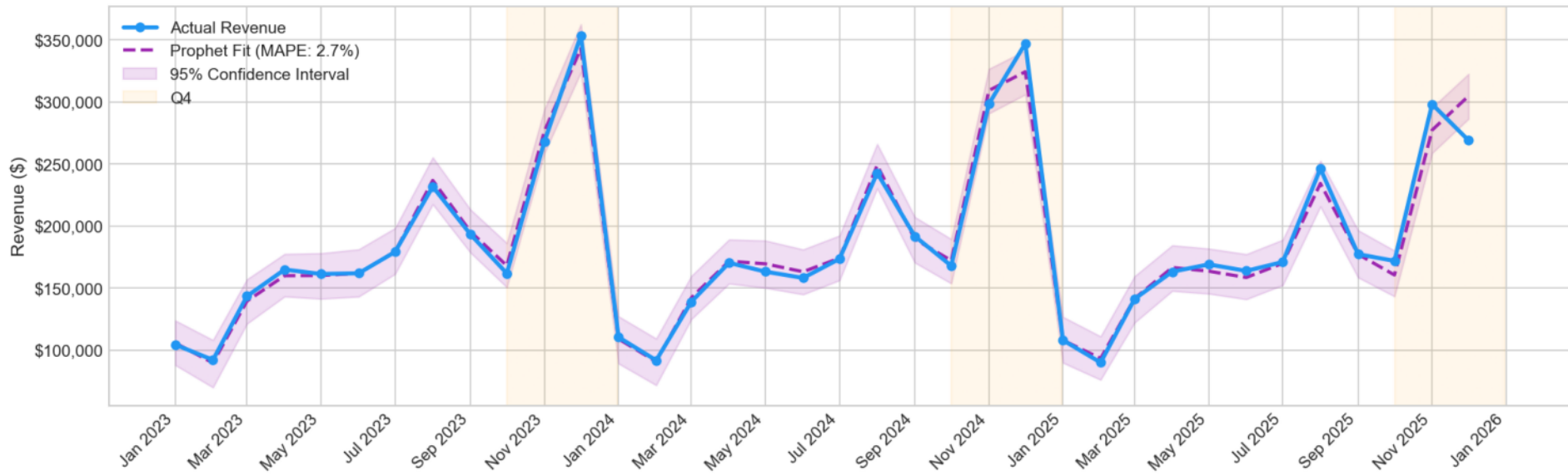


Forecast Residuals (Actual - Fitted)

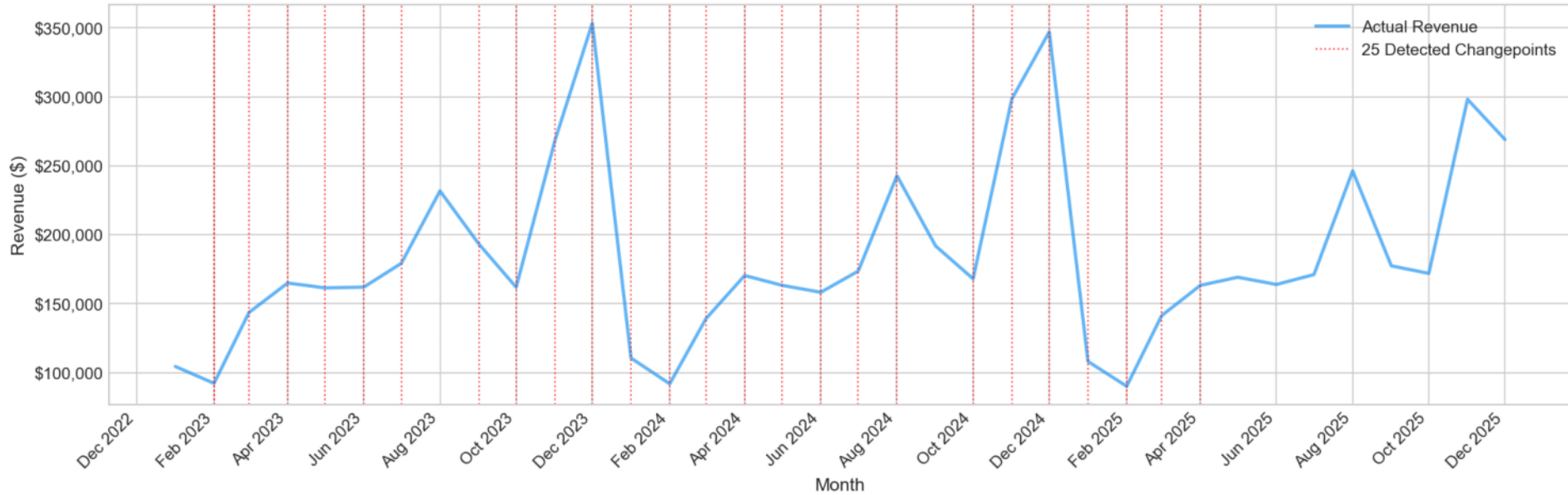


Model 3: Facebook Prophet — Historical Fit

Facebook Prophet — Historical Fit with Confidence Intervals



Prophet Changepoint Detection — Where Trend Shifted (25 changepoints detected)



12-Month Forward Forecast: Jan–Dec 2026

Northwind Apparel Co. — Revenue Forecast Jan–Dec 2026
Prophet Model | MAPE: 2.7% | 2026 Projected Annual: \$1,966,619

